Create the standard calculator application in android:

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:tools="http://schemas.android.com/tools">
  <application
    android:allowBackup="true"
    android:dataExtractionRules="@xml/data_extraction_rules"
    android:fullBackupContent="@xml/backup_rules"
    android:icon="@mipmap/ic_launcher"
    android:label="@string/app_name"
    android:roundIcon="@mipmap/ic_launcher_round"
    android:supportsRtl="true"
    android:theme="@style/Theme.Calculator">
    <activity
      android:name=".MainActivity"
      android:exported="true">
      <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
      </intent-filter>
    </activity>
  </application>
</manifest>
```

MainActivity.java

```
package com.example.calculator1;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
  private TextView tvResult;
  private String currentInput = "";
  private double firstValue = 0.0;
  private String operator = "";

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    tvResult = findViewById(R.id.tvResult);
    setOperatorListener(R.id.btnAdd, "+");
    setOperatorListener(R.id.btnSub, "-");
    setOperatorListener(R.id.btnMul, "*");
    setOperatorListener(R.id.btnDiv, "/");
    setOperatorListener(R.id.btnMod, "%");
    findViewById(R.id.btnClear).setOnClickListener(v -> {
      currentInput = "";
      firstValue = 0.0;
      operator = "";
      tvResult.setText("0");
    });
    findViewById(R.id.btnDel).setOnClickListener(v -> {
      if (currentInput.length() > 0) {
```

```
            currentInput = currentInput.substring(0, currentInput.length() - 1);
            tvResult.setText(currentInput.isEmpty() ? "0" : currentInput);
        }
    });
    findViewById(R.id.btnEqual).setOnClickListener(v -> calculateResult());
}

public void onDigitClick(View view) {
    Button btn = (Button) view;
    currentInput += btn.getText().toString();
    tvResult.setText(currentInput);
}

private void setOperatorListener(int btnId, String op) {
    findViewById(btnId).setOnClickListener(v -> {
        if (!currentInput.isEmpty()) {
            firstValue = Double.parseDouble(currentInput);
            operator = op;
            currentInput = "";          }
    });
}

private void calculateResult() {
    if (!currentInput.isEmpty() && !operator.isEmpty()) {
        double secondValue = Double.parseDouble(currentInput);
        double result = 0.0;
        switch (operator) {
            case "+":
                result = firstValue + secondValue;
                break;
            case "-":
                result = firstValue - secondValue;
                break;
            case "*":
                result = firstValue * secondValue;
                break;
            case "/":
                result = firstValue / secondValue;
                break;
            case "%":
                result = firstValue % secondValue;
                break;
        }
        tvResult.setText(String.valueOf(result));
        currentInput = String.valueOf(result);
        operator = "";
    }
}
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
```

```
    android:padding="10dp"
    android:background="#FFFFFF">
    <TextView
        android:id="@+id/tvResult"
        android:layout_width="match_parent"
        android:layout_height="250dp"
        android:background="#EEEEEE"
        android:gravity="bottom|end"
        android:padding="50dp"
        android:text="0"
        android:textColor="#000000"
        android:textSize="48sp" />
    <GridLayout
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:columnCount="4"
        android:rowCount="5"
        android:layout_marginTop="10dp"
        android:alignmentMode="alignMargins">
        <Button android:id="@+id/btnDel" android:text="DEL" style="@style/CalcBtn"/>
        <Button android:id="@+id/btnClear" android:text="C" style="@style/CalcBtn"/>
        <Button android:id="@+id/btnMod" android:text="%" style="@style/CalcBtn"/>
        <Button android:id="@+id/btnAdd" android:text="+" style="@style/CalcBtn"/>
        <Button android:text="1" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:text="2" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:text="3" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:id="@+id/btnDiv" android:text="/" style="@style/CalcBtn"/>
        <Button android:text="4" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:text="5" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:text="6" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:id="@+id/btnSub" android:text="-" style="@style/CalcBtn"/>
        <Button android:text="7" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:text="8" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:text="9" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:id="@+id/btnMul" android:text="X" style="@style/CalcBtn"/>
        <Button android:text="." android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:text="0" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button
            android:id="@+id/btnEqual"
            android:text="="
            style="@style/CalcBtn"
            android:layout_columnSpan="2"
            android:layout_gravity="fill_horizontal"/>
    </GridLayout>
</LinearLayout>
```

Colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="teal_primary">#009688</color>
    <color name="teal_dark">#00796B</color>
    <color name="button_gray">#E0E0E0</color>
</resources>
```

Themes.xml

```
<resources xmlns:tools="http://schemas.android.com/tools">
<style name="Base.Theme.Calculator" parent="Theme.AppCompat.Light.DarkActionBar">
  <item name="colorPrimary">@color/teal_primary</item>
  <item name="colorPrimaryDark">@color/teal_dark</item>
  <item name="colorAccent">@color/teal_primary</item>
  <item name="titleTextColor">@color/white</item>
</style>
<style name="Theme.Calculator" parent="Base.Theme.Calculator" />
<style name="CalcBtn">
  <item name="android:layout_width">0dp</item>
  <item name="android:layout_height">0dp</item>
  <item name="android:layout_columnWeight">1</item>
  <item name="android:layout_rowWeight">1</item>
  <item name="android:textSize">20sp</item>
  <item name="android:textStyle">bold</item>
  <item name="android:layout_margin">6dp</item>
  <item name="android:background">@color/button_gray</item>
  <item name="android:textColor">@color/black</item>
</style>
</resources>
```

2. Create a login screen with EditText fields for username and password, and a Button to submit. Include validation for empty fields.

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
  package="com.example.uielements">
  <application
     android:allowBackup="true"
     android:icon="@mipmap/ic_launcher"
     android:label="@string/app_name"
     android:roundIcon="@mipmap/ic_launcher_round"
     android:supportsRtl="true"
     android:theme="@style/Theme.UIElements">
     <activity android:name=".MainActivity"
        android:exported="true">
        <intent-filter>
           <action android:name="android.intent.action.MAIN" />
           <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
     </activity>
  </application>
</manifest>
```

MainActivity.java

```
package com.example.uielements;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
```

```
import androidx.appcompat.widget.Toolbar;
import androidx.fragment.app.FragmentTransaction;
public class MainActivity extends AppCompatActivity {
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    Toolbar toolbar = findViewById(R.id.toolbar);
    setSupportActionBar(toolbar);
    EditText et_user_name = findViewById(R.id.et_user_name);
    EditText et_password = findViewById(R.id.et_password);
    Button btnReset = findViewById(R.id.btn_reset);
    Button btnSubmit = findViewById(R.id.btn_submit);
    btnReset.setOnClickListener(v -> {
      et_user_name.setText("");
      et_password.setText("");
    });
    btnSubmit.setOnClickListener(v -> {
      String uname = et_user_name.getText().toString().trim();
      String pwd = et_password.getText().toString().trim();
      if (uname.isEmpty()) {
        et_user_name.setError("Username is required");
        et_user_name.requestFocus();
        return;        }
      if (pwd.isEmpty()) {
        et_password.setError("Password is required");
        et_password.requestFocus();
        return;        }
      Toast.makeText(MainActivity.this, "Welcome " + uname, Toast.LENGTH_LONG).show();
    });
  }}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity">
  <com.google.android.material.appbar.AppBarLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:theme="@style/ThemeOverlay.AppCompat.Dark.ActionBar">
    <androidx.appcompat.widget.Toolbar
      android:id="@+id/toolbar"
      android:layout_width="match_parent"
      android:layout_height="?attr/actionBarSize"
      app:title="UI Elements Demo"
      app:titleTextColor="@android:color/white"
      app:popupTheme="@style/ThemeOverlay.AppCompat.Light"/>
  </com.google.android.material.appbar.AppBarLayout>
  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
    android:orientation="vertical"
    android:padding="16dp"
    android:layout_marginTop="80dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="25sp"
        android:text="Login"
        android:layout_gravity="center_horizontal"/>
    <com.google.android.material.textfield.TextInputLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp">
        <com.google.android.material.textfield.TextInputEditText
            android:id="@+id/et_user_name"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Username"
            android:inputType="textPersonName" />
    </com.google.android.material.textfield.TextInputLayout>
    <com.google.android.material.textfield.TextInputLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        app:passwordToggleEnabled="true"> <com.google.android.material.textfield.TextInputEditText
        android:id="@+id/et_password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword" />
    </com.google.android.material.textfield.TextInputLayout>
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:orientation="horizontal"
        android:layout_marginTop="16dp">
        <Button
            android:id="@+id/btn_reset"
            android:text="Reset"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginEnd="16dp"/>
        <Button
            android:id="@+id/btn_submit"
            android:text="Submit"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>
    </LinearLayout>
    <FrameLayout
        android:id="@+id/fragment_container"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:layout_marginTop="16dp"/>
</LinearLayout>
```

```
</androidx.coordinatorlayout.widget.CoordinatorLayout>
```

3. Create an android application to display Alert Dialog on pressing the Back button.

MainActivity.java

```
package com.example.alertdialogapp;
import android.os.Bundle;
import androidx.activity.OnBackPressedCallback;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    OnBackPressedCallback callback = new OnBackPressedCallback(true) {
      @Override
      public void handleOnBackPressed() {
        showExitDialog();
      }
    };
    getOnBackPressedDispatcher().addCallback(this, callback);
  }
  private void showExitDialog() {
    new AlertDialog.Builder(this)
        .setTitle("Exit Application")
        .setMessage("Are you sure you want to exit?")
        .setCancelable(false) // Prevent closing by clicking outside
        .setPositiveButton("Yes", (dialog, which) -> {
          finish();
        })
        .setNegativeButton("No", (dialog, which) -> {
          dialog.dismiss();
        })
        .show();
  }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center"
  android:orientation="vertical">
  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Press the Back Button to see the Alert"
    android:textSize="18sp"
    android:textStyle="bold" />
</LinearLayout>
```

4. Create an android application to pass the data from one activity to another activity in the same application using intent.

Androidmanifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:tools="http://schemas.android.com/tools">

  <application
     android:allowBackup="true"
     android:dataExtractionRules="@xml/data_extraction_rules"
     android:fullBackupContent="@xml/backup_rules"
     android:icon="@mipmap/ic_launcher"
     android:label="@string/app_name"
     android:roundIcon="@mipmap/ic_launcher_round"
     android:supportsRtl="true"
     android:theme="@style/Theme.Datapassingapp">
     <activity
        android:name=".MainActivity"
        android:exported="true">
        <intent-filter>
           <action android:name="android.intent.action.MAIN" />

           <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
     </activity>
     <activity android:name=".SecondActivity" />
  </application>

</manifest>
```

MainActivity.java

```
package com.example.datapassingapp; // TODO: Check your package name
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
  // Declare variables
  private EditText etInput;
  private Button btnSend;
  @Override
  protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_main);
     // 1. Bind Views (These IDs match activity_main.xml)
     etInput = findViewById(R.id.etInput);
     btnSend = findViewById(R.id.btnSend);
     // 2. Set Button Click Listener
     btnSend.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
           // Get the string from the EditText
           String message = etInput.getText().toString();
```

```
            // Create the Intent object (From Here -> To SecondActivity)
            Intent intent = new Intent(MainActivity.this, SecondActivity.class);
            // Pass the data (Key: "USER_MESSAGE", Value: message)
            intent.putExtra("USER_MESSAGE", message);
            // Start the new activity
            startActivity(intent);
        }
    });
  }
}
```

SecondActivity.java

```
package com.example.datapassingapp; // TODO: Check your package name
import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
public class SecondActivity extends AppCompatActivity {
  private TextView tvResult;
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_second);
    // 1. Bind View (Matches activity_second.xml)
    tvResult = findViewById(R.id.tvResult);
    // 2. Get the Intent that started this activity
    Intent intent = getIntent();
    // 3. Extract the string using the same Key "USER_MESSAGE"
    if (intent != null) {
      String receivedText = intent.getStringExtra("USER_MESSAGE");
      // Set the text to the TextView
      tvResult.setText(receivedText);
    }
  }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:gravity="center"
  android:padding="20dp">
  <EditText
    android:id="@+id/etInput"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Type a message here"
    android:textSize="18sp" />
  <Button
    android:id="@+id/btnSend"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Send to Second Activity"
```

```
    android:layout_marginTop="20dp" />

</LinearLayout>
```

Activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:gravity="center"
  android:padding="20dp">
  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Received Data:"
    android:textSize="16sp" />
  <TextView
    android:id="@+id/tvResult"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Waiting..."
    android:textSize="24sp"
    android:textStyle="bold"
    android:layout_marginTop="10dp"
    android:textColor="@android:color/holo_blue_dark"/>
</LinearLayout>
```

5. Create a user interface with a TextView, Button, and ImageView. When the button is clicked, the image should be changed dynamically.

MainActivity.java

```
package com.example.imagechanger; // TODO: Check your package name
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
  // Declare View variables
  private TextView tvStatus;
  private ImageView ivDisplay;
  private Button btnChange;
  // Flag to track which image is currently showing
  private boolean isImageOne = true;
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    // 1. Initialize Views
    tvStatus = findViewById(R.id.tvStatus);
    ivDisplay = findViewById(R.id.ivDisplay);
    btnChange = findViewById(R.id.btnChange);
    // 2. Set Button Click Listener
    btnChange.setOnClickListener(new View.OnClickListener() {
```

```
        @Override
        public void onClick(View v) {
            if (isImageOne) {
                // Switch to second image
                ivDisplay.setImageResource(R.drawable.image_two);
                tvStatus.setText("Current Image: Two");
                isImageOne = false; // Update flag
            } else {
                // Switch back to first image
                ivDisplay.setImageResource(R.drawable.image_one);
                tvStatus.setText("Current Image: One");
                isImageOne = true; // Update flag
            }
        }
    });
  }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:gravity="center"
  android:padding="20dp">
  <TextView
    android:id="@+id/tvStatus"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Current Image: One"
    android:textSize="22sp"
    android:textStyle="bold"
    android:layout_marginBottom="20dp"/>
  <ImageView
    android:id="@+id/ivDisplay"
    android:layout_width="250dp"
    android:layout_height="250dp"
    android:src="@drawable/image_one"
    android:scaleType="centerCrop"
    android:layout_marginBottom="30dp"/>
  <Button
    android:id="@+id/btnChange"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Change Image"
    android:padding="15dp"/>
</LinearLayout>
```

6. Create the background service android application to play the ringtone/music.

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:tools="http://schemas.android.com/tools">

  <application
```

```
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Backgroundmusic">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <service android:name=".MusicService" />
    </application>

</manifest>
```

MainActivity.java

```
package com.example.backgroundmusic; // TODO: Check your package name
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    Button btnStart, btnStop;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnStart = findViewById(R.id.btnStart);
        btnStop = findViewById(R.id.btnStop);
        // Listener to Start Service
        btnStart.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent serviceIntent = new Intent(MainActivity.this, MusicService.class);
                startService(serviceIntent);
            }
        });
        // Listener to Stop Service
        btnStop.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent serviceIntent = new Intent(MainActivity.this, MusicService.class);
                stopService(serviceIntent);
            }
        });
    }
}
```

MusicService.java

```
package com.example.backgroundmusic; // TODO: Check your package name
import android.app.Service;
import android.content.Intent;
import android.media.MediaPlayer;
import android.os.IBinder;
import android.provider.Settings;
import android.widget.Toast;
import androidx.annotation.Nullable;
public class MusicService extends Service {
  private MediaPlayer player;
  @Override
  @Nullable
  public IBinder onBind(Intent intent) {
     // We are not binding to an activity, so return null
     return null;
  }
  @Override
  public int onStartCommand(Intent intent, int flags, int startId) {
     // 1. Create the MediaPlayer logic
     if (player == null) {
        // Use the default system ringtone
        player = MediaPlayer.create(this, Settings.System.DEFAULT_RINGTONE_URI);
        player.setLooping(true); // Loop the music
     }
     // 2. Start the player if it's not already playing
     if (!player.isPlaying()) {
        player.start();
        Toast.makeText(this, "Music Service Started", Toast.LENGTH_SHORT).show();
     }
     // START_STICKY ensures the service restarts if the system kills it
     return START_STICKY;
  }
  @Override
  public void onDestroy() {
     super.onDestroy();
     // 3. Cleanup: Stop and release the player when service is stopped
     if (player != null) {
        if (player.isPlaying()) {
           player.stop();
        }
        player.release();
        player = null;
     }
     Toast.makeText(this, "Music Service Stopped", Toast.LENGTH_SHORT).show();
  }
}
```

7.Create an android application which automatically notify the user when Aeroplane mode is turned on or off using a broadcast receiver.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
```

```
  android:gravity="center"
  android:padding="20dp">
  <TextView
     android:id="@+id/tvInstruction"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Toggle Airplane Mode via your Quick Settings Panel to test."
     android:textSize="18sp"
     android:gravity="center"
     android:layout_marginBottom="20dp"/>
  <TextView
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="(Toast notification will appear)"
     android:textSize="14sp"
     android:textColor="#666666"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.airplanemodeapp; // TODO: Check your package name
import android.content.Intent;
import android.content.IntentFilter;
import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
  private AirplaneModeReceiver airplaneModeReceiver;
  @Override
  protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_main);
     // Initialize the receiver
     airplaneModeReceiver = new AirplaneModeReceiver();
  }
  @Override
  protected void onStart() {
     super.onStart();
     // 1. Create an IntentFilter for Airplane Mode
     IntentFilter filter = new IntentFilter(Intent.ACTION_AIRPLANE_MODE_CHANGED);
     // 2. Register the receiver dynamically
     // This is required for Android 8.0 (Oreo) and above
     registerReceiver(airplaneModeReceiver, filter);
  }
  @Override
  protected void onStop() {
     super.onStop();
     // 3. Unregister the receiver to save battery and prevent crashes
     if (airplaneModeReceiver != null) {
        unregisterReceiver(airplaneModeReceiver);
     }
  }
}
```

AirplaneModeReceiver.java

```
package com.example.airplanemodeapp;
import android.content.BroadcastReceiver;
```

```
import android.content.Context;
import android.content.Intent;
import android.os.CountDownTimer;
import android.widget.Toast;
public class AirplaneModeReceiver extends BroadcastReceiver {
  @Override
  public void onReceive(Context context, Intent intent) {
     if (Intent.ACTION_AIRPLANE_MODE_CHANGED.equals(intent.getAction())) {
        boolean isAirplaneModeOn = intent.getBooleanExtra("state", false);
        String message = isAirplaneModeOn ? "Airplane Mode is ON " : "Airplane Mode is OFF ";
        showCustomDurationToast(context, message, 10000);
     }
  }
  private void showCustomDurationToast(Context context, String message, int durationInMillis) {
     final Toast toast = Toast.makeText(context, message, Toast.LENGTH_SHORT);
     new CountDownTimer(durationInMillis, 1000) {

        public void onTick(long millisUntilFinished) {
           toast.show();
        }
        public void onFinish() {
           // Ensure the toast disappears when time is up
           toast.cancel();
        }
     }.start();
  }
}
```

8.Explain the Android Activity life cycle in detail. Implement a simple app that logs the lifecycle methods (onCreate(), onStart(), onResume(), onPause(), onStop(), onDestroy())

MainActivity.java

```
package com.example.lifecyclelogger; // TODO: Check your package name
import android.os.Bundle;
import android.util.Log;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
  // Tag for filtering logs in Android Studio
  private static final String TAG = "LifecycleEvent";
  @Override
  protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_main);

     Log.d(TAG, "onCreate() called: Activity Initialized");
  }
  @Override
  protected void onStart() {
     super.onStart();
     Log.d(TAG, "onStart() called: Activity is becoming visible");
  }
  @Override
  protected void onResume() {
     super.onResume();
```

```
        Log.d(TAG, "onResume() called: Activity is interactive (Top of stack)");
    }
    @Override
    protected void onPause() {
        super.onPause();
        Log.d(TAG, "onPause() called: User is leaving (or app partially obscured)");
    }
    @Override
    protected void onStop() {
        super.onStop();
        Log.d(TAG, "onStop() called: Activity is no longer visible");
    }
    @Override
    protected void onRestart() {
        super.onRestart();
        Log.d(TAG, "onRestart() called: Activity returning from stopped state");
    }
    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d(TAG, "onDestroy() called: Activity is being destroyed");
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Check Logcat for 'LifecycleEvent'"
        android:textSize="20sp"
        android:textStyle="bold" />
</LinearLayout>
```

9. Create a simple Hello World application in Android Studio. Explain the purpose of the MainActivity and the activity_main.xml file.

MainActivity.java

```
package com.example.helloworld;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  android:layout_width="match_parent"
  android:layout_height="match_parent">

  <TextView
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Hello World!"
     app:layout_constraintTop_toTopOf="parent"
     app:layout_constraintBottom_toBottomOf="parent"
     app:layout_constraintStart_toStartOf="parent"
     app:layout_constraintEnd_toEndOf="parent"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

10 . Create an android application to demonstrate the following event listeners on the following widgets:
Button: on click listener, long click listener
Image: Touch listener with these motion events :Action up, action down and action pointer down
Edit Text : key listener on Enter key
The Appropriate message should be displayed in text view.

MainActivity.java

```
package com.example.eventlistenersdemo;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.KeyEvent;
import android.view.MotionEvent;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ImageView;
import android.widget.TextView;
public class MainActivity extends AppCompatActivity {
  Button btnClick;
  ImageView imageView;
  EditText editText;
  TextView textView;
  @Override
  protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_main);
     btnClick = findViewById(R.id.btnClick);
     imageView = findViewById(R.id.imageView);
     editText = findViewById(R.id.editText);
     textView = findViewById(R.id.textView);
     btnClick.setOnClickListener(v ->
           textView.setText("Button Clicked"));
     btnClick.setOnLongClickListener(v -> {
        textView.setText("Button Long Pressed");
        return true;
```

```
        });
        imageView.setOnTouchListener((v, event) -> {
            switch (event.getAction()) {
                case MotionEvent.ACTION_DOWN:
                    textView.setText("Image Action Down");
                    break;
                case MotionEvent.ACTION_UP:
                    textView.setText("Image Action Up");
                    break;
                case MotionEvent.ACTION_POINTER_DOWN:
                    textView.setText("Pointer Down (Multiple Fingers)");
                    break;
            }
            return true;
        });
        editText.setOnKeyListener((v, keyCode, event) -> {
            if (event.getAction() == KeyEvent.ACTION_DOWN &&
                    keyCode == KeyEvent.KEYCODE_ENTER) {
                textView.setText("Enter Key Pressed");
                return true;
            }
            return false;
        });
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <ImageView
        android:id="@+id/imageView"
        android:layout_width="100dp"
        android:layout_height="100dp"
        android:src="@mipmap/ic_launcher"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent" />
    <EditText
        android:id="@+id/editText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:hint="Press Enter Key" />
    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Event Message Will Appear Here"
        android:textSize="18sp" />
    <Button
```

```
    android:id="@+id/btnClick"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Click or Long Press Me" />
</LinearLayout>
```

11 . Implement a custom color scheme for your registration form. Set the background color of the entire layout to a light color, and use a contrasting color for the Submit button.

colors.xml

```
<resources>
  <color name="light_bg">#B9FF00</color>
  <color name="submit_color">#000000</color>
  <color name="white">#FFFFFF</color>
</resources>
```

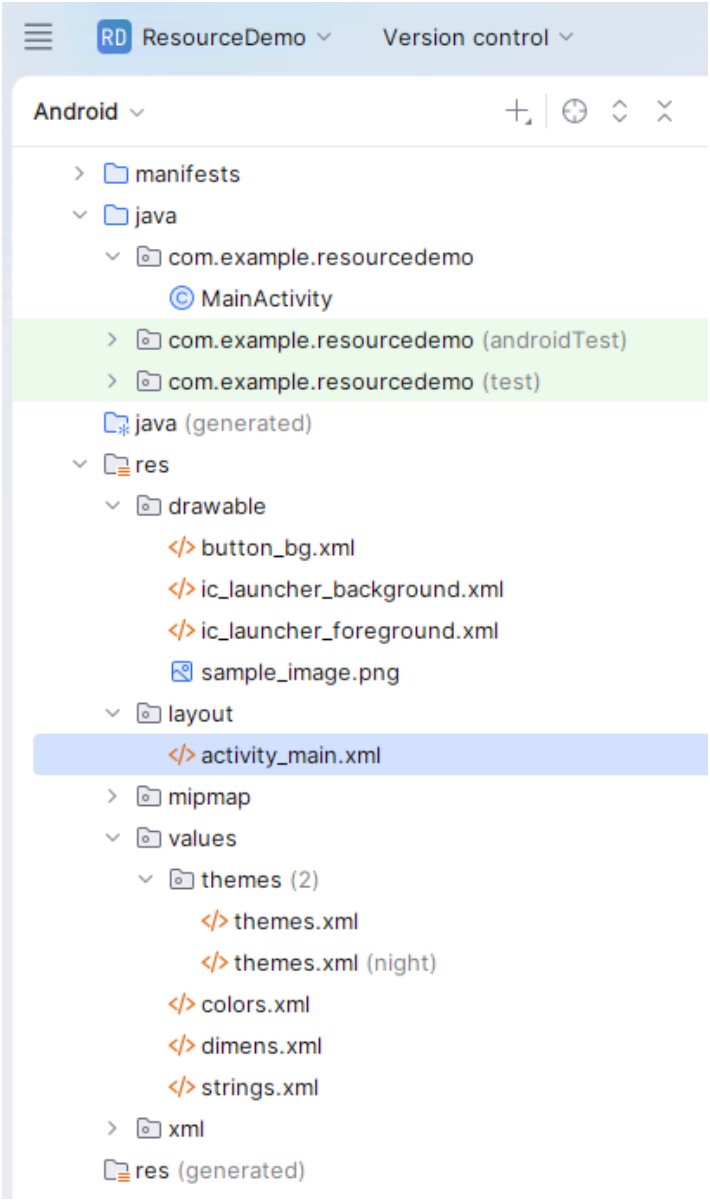
activity_main

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity">

  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:background="@color/light_bg">

    <EditText
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Name" />

    <EditText
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Email"
      android:layout_marginTop="10dp" />

    <Button
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Submit"
      android:textColor="@color/white"
      android:backgroundTint="@color/submit_color"
      android:layout_marginTop="20dp" />
  </LinearLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```

12. Insert the new contents in the following resources and demonstrate their uses in the android application Android Resources: (Color, Theme, String, Drawable, Dimension, Image).

Project Name: ResourceDemo

File Structure:



MainActivity.java

```
package com.example.resourcedemo;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class MainActivity extends AppCompatActivity {
  @Override
  protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_main);
  }
```

```
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:orientation="vertical"
  android:padding="@dimen/padding"
  android:layout_width="match_parent"
  android:layout_height="match_parent">

  <TextView
     android:text="@string/welcome_text"
     android:textSize="@dimen/text_size"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"/>

  <ImageView
     android:src="@drawable/sample_image"
     android:layout_width="@dimen/image_size"
     android:layout_height="@dimen/image_size"
     android:layout_marginTop="20dp"/>

  <Button
     android:text="@string/button_text"
     android:textColor="@color/white"
     android:background="@drawable/button_bg"
     android:layout_width="match_parent"
     android:layout_height="wrap_content"
     android:layout_marginTop="20dp"/>
</LinearLayout>
```

Strings.xml

```
<resources>
  <string name="app_name">ResourceDemo</string>
  <string name="welcome_text">Welcome to Resource Demo</string>
  <string name="button_text">Submit</string>
</resources>
```

Colors.xml

```
<resources>
  <color name="bg_color">#F0F0F0</color>
  <color name="primary_color">#6200EE</color>
  <color name="white">#FFFFFF</color>
</resources>
```

app-> res -> drawable-> button_bg.xml

Button_bg.xml

```
<shape xmlns:android="http://schemas.android.com/apk/res/android">
  <solid android:color="@color/primary_color"/>
  <corners android:radius="12dp"/>
</shape>
```

app->res->values-> dimens.xml

Dimens.xml

```
<resources>
  <dimen name="padding">20dp</dimen>
  <dimen name="text_size">20sp</dimen>
  <dimen name="image_size">150dp</dimen>
</resources>
```

Themes.xml

```
<resources xmlns:tools="http://schemas.android.com/tools">
  <style name="Theme.ResourceDemo" parent="Theme.Material3.DayNight.NoActionBar">
     <item name="android:windowBackground">@color/bg_color</item>
  </style>
</resources>
```

13. .Define the following string resources in the strings.xml file for the registration form: Full Name, Email Password Confirm Password and Submit
Use these string resources in the corresponding TextView and Button in your layout XML.

Project Name: RegistrationForm

MainActivity.java

```
package com.example.registrationform;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_main);
  }
}
```

Strings.xml

```
<resources>
  <string name="app_name">RegistrationForm</string>
  <string name="full_name">Full Name</string>
  <string name="email">Email</string>
  <string name="password">Password</string>
  <string name="confirm_password">Confirm Password</string>
  <string name="submit">Submit</string>
</resources>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:orientation="vertical"
  android:padding="20dp"
```

```
android:layout_width="match_parent"
android:layout_height="match_parent">

<TextView
    android:text="@string/full_name"
    android:textSize="18sp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

<EditText
    android:hint="@string/full_name"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<TextView
    android:text="@string/email"
    android:textSize="18sp"
    android:layout_marginTop="10dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

<EditText
    android:hint="@string/email"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<TextView
    android:text="@string/password"
    android:textSize="18sp"
    android:layout_marginTop="10dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

<EditText
    android:hint="@string/password"
    android:inputType="textPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<TextView
    android:text="@string/confirm_password"
    android:textSize="18sp"
    android:layout_marginTop="10dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

<EditText
    android:hint="@string/confirm_password"
    android:inputType="textPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<Button
    android:text="@string/submit"
    android:layout_marginTop="20dp"
    android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"/>

</LinearLayout>
```

14. Create a registration form where the form fields (Name, Email, Password, Confirm Password) are stacked vertically. Include a Submit button at the bottom of the form.

Project Name: RegistrationForm

MainActivity.java

```
package com.example.registrationform;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {
  @Override
  protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
  }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:orientation="vertical"
  android:padding="24dp"
  android:layout_width="match_parent"
  android:layout_height="match_parent">

  <EditText
      android:hint="@string/full_name"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"/>

  <EditText
      android:hint="@string/email"
      android:inputType="textEmailAddress"
      android:layout_marginTop="12dp"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"/>

  <EditText
      android:hint="@string/password"
      android:inputType="textPassword"
      android:layout_marginTop="12dp"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"/>

  <EditText
      android:hint="@string/confirm_password"
      android:inputType="textPassword"
      android:layout_marginTop="12dp"
      android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"/>

    <Button
        android:text="@string/submit"
        android:layout_marginTop="20dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

Strings.xml

```
<resources>
    <string name="app_name">RegistrationForm</string>
    <string name="full_name">Full Name</string>
    <string name="email">Email</string>
    <string name="password">Password</string>
    <string name="confirm_password">Confirm Password</string>
    <string name="submit">Submit</string>
</resources>
```

15. Create an android application, with one button and image view also set the background image in the application, the image should be changed on each click of the button (Use two images).

Project Name: ImageSwitcherApp

Add Images

app → res → drawable

Add 3 images:

bg_image.jpg (Background)

image1.png

image2.png

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:background="@drawable/bg_image"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ImageView
        android:id="@+id/imageView"
        android:src="@drawable/image1"
        android:layout_width="200dp"
        android:layout_height="200dp"/>

    <Button
        android:id="@+id/button"
        android:text="Change Image"
```

```
        android:layout_marginTop="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.imageswitcherapp;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Button;
import android.widget.ImageView;
public class MainActivity extends AppCompatActivity {

    ImageView imageView;
    Button button;

    boolean isFirstImage = true;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        imageView = findViewById(R.id.imageView);
        button = findViewById(R.id.button);

        button.setOnClickListener(v -> {

            if(isFirstImage){
                imageView.setImageResource(R.drawable.image2);
                isFirstImage = false;
            }
            else{
                imageView.setImageResource(R.drawable.image1);
                isFirstImage = true;
            }

        });
    }
}
```

Create an android application to demonstrate the use of sub menu the toast should be appeared by selecting the sub menu item.

menu_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/menu_options"
        android:title="Options">

        <menu>
            <item
                android:id="@+id/sub_item1"
                android:title="Profile"/>
```

```
        <item
            android:id="@+id/sub_item2"
            android:title="Settings"/>

        <item
            android:id="@+id/sub_item3"
            android:title="Logout"/>
    </menu>

  </item>

</menu>
```

MainActivity.java

```
package com.example.submenudemo;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
  }

  @Override
  public boolean onCreateOptionsMenu(Menu menu) {
    getMenuInflater().inflate(R.menu.menu_main, menu);
    return true;
  }

  @Override
  public boolean onOptionsItemSelected(MenuItem item) {

    int id = item.getItemId();

    if(id == R.id.sub_item1){
        Toast.makeText(this, "Profile Selected", Toast.LENGTH_SHORT).show();
        return true;
    }
    else if(id == R.id.sub_item2){
        Toast.makeText(this, "Settings Selected", Toast.LENGTH_SHORT).show();
        return true;
    }
    else if(id == R.id.sub_item3){
        Toast.makeText(this, "Logout Selected", Toast.LENGTH_SHORT).show();
        return true;
    }

    return super.onOptionsItemSelected(item);
  }
```

```
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:gravity="center"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

</LinearLayout>
```

themes.xml

```
<resources xmlns:tools="http://schemas.android.com/tools">
   <!-- Base application theme. -->
   <style name="Base.Theme.SubMenuDemo" parent="Theme.Material3.DayNight">
      <!-- Customize your light theme here. -->
      <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
   </style>

   <style name="Theme.SubMenuDemo" parent="Base.Theme.SubMenuDemo" />
</resources>
```

Design a form with a RadioGroup for selecting the gender (Male/Female/Other). Include a Submit button that shows the selected gender in a TextView.

MainActivity.java

```
package com.example.radiogroup;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Button;
import android.widget.RadioGroup;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

   RadioGroup radioGroup;
   Button buttonSubmit;
   TextView textViewResult;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      radioGroup = findViewById(R.id.radioGroup);
      buttonSubmit = findViewById(R.id.buttonSubmit);
      textViewResult = findViewById(R.id.textViewResult);

      buttonSubmit.setOnClickListener(v -> {

         int selectedId = radioGroup.getCheckedRadioButtonId();
```

```
        if(selectedId == -1){
            textViewResult.setText("Please select a gender");
        }
        else{
            if(selectedId == R.id.radioMale){
                textViewResult.setText("Selected Gender: Male");
            }
            else if(selectedId == R.id.radioFemale){
                textViewResult.setText("Selected Gender: Female");
            }
            else{
                textViewResult.setText("Selected Gender: Other");
            }
        }
    });
  }
}
```

strings.xml

```
<resources>
  <string name="app_name">GenderForm</string>
  <string name="select_gender">Select Gender</string>
  <string name="male">Male</string>
  <string name="female">Female</string>
  <string name="other">Other</string>
  <string name="submit">Submit</string>
</resources>
```

activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:orientation="vertical"
  android:padding="20dp"
  android:layout_width="match_parent"
  android:layout_height="match_parent">

  <TextView
    android:text="@string/select_gender"
    android:textSize="18sp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

  <RadioGroup
    android:id="@+id/radioGroup"
    android:layout_marginTop="10dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content">

    <RadioButton
      android:id="@+id/radioMale"
      android:text="@string/male"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"/>
    <RadioButton
      android:id="@+id/radioFemale"
      android:text="@string/female"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
    <RadioButton
        android:id="@+id/radioOther"
        android:text="@string/other"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

  </RadioGroup>

  <Button
    android:id="@+id/buttonSubmit"
    android:text="@string/submit"
    android:layout_marginTop="20dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

  <TextView
    android:id="@+id/textViewResult"
    android:textSize="18sp"
    android:layout_marginTop="20dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

</LinearLayout>
```

Create a RelativeLayout for a profile screen with:A Profile Image centered at the top of the screen.
A TextView displaying the name below the profile image, aligned to the center.
A Button below the name, aligned to the center.

string.xml

```
<resources>
   <string name="app_name">ProfileScreen</string>
   <string name="user_name">MEET PALAK</string>
   <string name="follow">Follow</string>
</resources>
```

MainActivity.java

```
package com.example.profilescreen;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {
   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);
   }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">

    <ImageView
        android:id="@+id/profileImage"
        android:src="@drawable/profile_image"
        android:layout_width="150dp"
        android:layout_height="150dp"
        android:layout_centerHorizontal="true"
        android:layout_alignParentTop="true"/>

    <TextView
        android:id="@+id/textName"
        android:text="@string/user_name"
        android:textSize="22sp"
        android:layout_below="@id/profileImage"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/buttonFollow"
        android:text="@string/follow"
        android:layout_below="@id/textName"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</RelativeLayout>
```

Write a program that logs the different lifecycle methods of an Activity (e.g., onCreate(), onStart(), onResume(), onPause(), onStop(), onDestroy()) to the logout.

MainActivity.java

```
package com.example.activitylifecycledemo;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.util.Log;

public class MainActivity extends AppCompatActivity {

    private static final String TAG = "LifecycleDemo";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Log.d(TAG, "onCreate() called");
    }

    @Override
    protected void onStart() {
```

```
        super.onStart();
        Log.d(TAG, "onStart() called");
    }

    @Override
    protected void onResume() {
        super.onResume();
        Log.d(TAG, "onResume() called");
    }

    @Override
    protected void onPause() {
        super.onPause();
        Log.d(TAG, "onPause() called");
    }

    @Override
    protected void onStop() {
        super.onStop();
        Log.d(TAG, "onStop() called");
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d(TAG, "onDestroy() called");
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:text="Check Logcat for Lifecycle Methods"
        android:textSize="18sp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

Design a ListView that displays a list of items. Each item should be a simple TextView with a string from an array of names. The list should be populated dynamically in your Activity.

MainActivity.java

```
package com.example.listviewdemo;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;

public class MainActivity extends AppCompatActivity {
```

```
    ListView listView;

    String[] names = {
            " ",
            "MEET",
            "PALAK",
            "DEEPIKA",
            "PRATHAM",
            "PARTH",
            "VAIBHAV",
            "ARYAN",
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);

        ArrayAdapter<String> adapter =
                new ArrayAdapter<>(this,
                        android.R.layout.simple_list_item_1,
                        names);

        listView.setAdapter(adapter);
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ListView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/listView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>
```

Create an Activity that hosts a Fragment. When a button in the Activity is clicked, pass a string value to the Fragment and display it in a TextView.

res → layout → fragment_my.xml

fragment_my.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <TextView
        android:id="@+id/textViewFragment"
        android:text="Message will appear here"
```

```
        android:textSize="20sp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</FrameLayout>
```

java package → New → Fragment → Fragment (Blank) Name it: `MyFragment`

MyFragment.java

```
package com.example.fragmentdatademo;

import android.os.Bundle;
import androidx.fragment.app.Fragment;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;

public class MyFragment extends Fragment {

   TextView textView;

   @Override
   public View onCreateView(LayoutInflater inflater, ViewGroup container,
                            Bundle savedInstanceState) {

       View view = inflater.inflate(R.layout.fragment_my, container, false);

       textView = view.findViewById(R.id.textViewFragment);

       // Receive data from Activity
       if(getArguments() != null){
           String message = getArguments().getString("data");
           textView.setText(message);
       }

       return view;
   }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:padding="20dp"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <FrameLayout
       android:id="@+id/fragmentContainer"
       android:layout_width="match_parent"
       android:layout_height="460dp"
       android:layout_marginTop="20dp">

   </FrameLayout>
```

```
    <Button
        android:id="@+id/buttonSend"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Data to Fragment" />

</LinearLayout>
```

MainActivity.java

```
package com.example.fragmentdatademo;

import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentTransaction;

import android.os.Bundle;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    Button buttonSend;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        buttonSend = findViewById(R.id.buttonSend);

        buttonSend.setOnClickListener(v -> {

            MyFragment fragment = new MyFragment();

            Bundle bundle = new Bundle();
            bundle.putString("data", "Hello from 4021,5059,5081!");

            fragment.setArguments(bundle);

            FragmentManager fm = getSupportFragmentManager();
            FragmentTransaction ft = fm.beginTransaction();

            ft.replace(R.id.fragmentContainer, fragment);
            ft.commit();
        });
    }
}
```

Create an android application using linear layout and insert 10 animals in the list view and display the appropriate Toast.

Project Name: AnimalListApp

Activity_main.xml

```
 <?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
```

```
   android:padding="10dp"
   android:layout_width="match_parent"
   android:layout_height="match_parent">
   <ListView
       android:id="@+id/listViewAnimals"
       android:layout_width="match_parent"
       android:layout_height="match_parent"/>
</LinearLayout>
```

Mainactivity.java

```
package com.example.animallistapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import android.widget.AdapterView;
public class MainActivity extends AppCompatActivity {
   ListView listView;
   String[] animals = {
           "Lion",
           "Tiger",
           "Elephant",
           "Dog",
           "Cat",
           "Horse",
           "Monkey",
           "Zebra",
           "Bear",
           "Deer"
   };
   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);
       listView = findViewById(R.id.listViewAnimals);
       ArrayAdapter<String> adapter =
               new ArrayAdapter<>(this,
                       android.R.layout.simple_list_item_1,
                       animals);
       listView.setAdapter(adapter);
       listView.setOnItemClickListener((parent, view, position, id) -> {
           String selectedAnimal = animals[position];
           Toast.makeText(
                   MainActivity.this,
                   "Selected: " + selectedAnimal,
                   Toast.LENGTH_SHORT
           ).show();
```

```
        });
    }
}
```

Create an app that displays an ImageView. When the user clicks the ImageView, show a Toast with the message & Image clicked!

Project Name : ImageClickApp

Add one image “sample_image.png”

app-> res->drawable

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:gravity="center"
   android:layout_width="match_parent"
   android:layout_height="match_parent">
   <ImageView
       android:id="@+id/imageView"
       android:src="@drawable/sample_image"
       android:layout_width="200dp"
       android:layout_height="200dp"
       android:clickable="true"/>
</LinearLayout>
```

Mainactivity.java

```
package com.example.imageclickapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.ImageView;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
   ImageView imageView;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);
       imageView = findViewById(R.id.imageView);
       imageView.setOnClickListener(v -> {
           Toast.makeText(
                   MainActivity.this,
                   "Image clicked!",
                   Toast.LENGTH_SHORT
           ).show();
       }); }}
```

Create the background service android application to play the ringtone/music.

Project name: MusicServiceApp

Inside app-> res

Create a new folder “raw”

Paste any audio inside raw folder “music.mp3”

Create a java class -> “MusicService”

MusicService.java

```
package com.example.musicserviceapp;
import android.app.Service;
import android.content.Intent;
import android.media.MediaPlayer;
import android.os.IBinder;
public class MusicService extends Service {
   MediaPlayer mediaPlayer;
   @Override
   public void onCreate() {
       super.onCreate();
       mediaPlayer = MediaPlayer.create(this, R.raw.music);
       mediaPlayer.setLooping(true); // plays continuously
   }
   @Override
   public int onStartCommand(Intent intent, int flags, int startId) {
       mediaPlayer.start(); // start music
       return START_STICKY; // keeps service running
   }
   @Override
   public void onDestroy() {
       super.onDestroy();
       mediaPlayer.stop();
       mediaPlayer.release();
   }
   @Override
   public IBinder onBind(Intent intent) {
       return null;
   }
}
```

MainActivity.java

```
package com.example.musicserviceapp;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
public class MainActivity extends AppCompatActivity {
   Button startBtn, stopBtn;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);
       startBtn = findViewById(R.id.buttonStart);
       stopBtn = findViewById(R.id.buttonStop);
       startBtn.setOnClickListener(v -> {
           startService(new Intent(MainActivity.this, MusicService.class));
       });
       stopBtn.setOnClickListener(v -> {
```

```
            stopService(new Intent(MainActivity.this, MusicService.class));
        });
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:gravity="center"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <Button
        android:id="@+id/buttonStart"
        android:text="Start Music"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/buttonStop"
        android:text="Stop Music"
        android:layout_marginTop="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.MusicServiceApp">
        <service android:name=".MusicService"/>
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

Design a simple login form using ConstraintLayout that has:
Username and Password fields aligned vertically.
Login button centered horizontally below the fields.
Forgot Password? link aligned to the right of the password field.
Task: Position all the elements using constraints in the ConstraintLayout.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:padding="24dp">
   <EditText
       android:id="@+id/editUsername"
       android:hint="@string/username"
       android:layout_width="0dp"
       android:layout_height="wrap_content"
       app:layout_constraintTop_toTopOf="parent"
       app:layout_constraintStart_toStartOf="parent"
       app:layout_constraintEnd_toEndOf="parent"/>
   <EditText
       android:id="@+id/editPassword"
       android:hint="@string/password"
       android:inputType="textPassword"
       android:layout_width="0dp"
       android:layout_height="wrap_content"
       android:layout_marginTop="16dp"
       app:layout_constraintTop_toBottomOf="@id/editUsername"
       app:layout_constraintStart_toStartOf="parent"
       app:layout_constraintEnd_toEndOf="parent"/>
   <TextView
       android:id="@+id/textForgot"
       android:text="@string/forgot_password"
       android:textColor="@android:color/holo_blue_dark"
       android:layout_marginTop="8dp"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       app:layout_constraintTop_toBottomOf="@id/editPassword"
       app:layout_constraintEnd_toEndOf="@id/editPassword"/>
   <Button
       android:id="@+id/buttonLogin"
       android:text="@string/login"
       android:layout_marginTop="24dp"
       android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        app:layout_constraintTop_toBottomOf="@id/textForgot"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.loginformapp;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class MainActivity extends AppCompatActivity {
   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);
   }
}
```

Strings.xml

```
<resources>
   <string name="app_name">LoginFormApp</string>
   <string name="username">Username</string>
   <string name="password">Password</string>
   <string name="login">Login</string>
   <string name="forgot_password">Forgot Password?</string>
</resources>
```

Create an android application to demonstrate the use of sub menu the toast should be appeared by selecting the sub menu item

```
Name: SubMenuApp
In res → values → themes.xml
Make sure this line is correct
<style name="Base.Theme.SubMenuApp" parent="Theme.Material3.DayNight">
  </style>
```

Inside res folder create a menu folder

Inside menu-> menu_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
   <item
       android:id="@+id/menu_more"
       android:title="More Options">
       <menu>
           <item
               android:id="@+id/profile"
               android:title="Profile"/>
           <item
               android:id="@+id/settings"
```

```
                android:title="Settings"/>
            <item
                android:id="@+id/logout"
                android:title="Logout"/>
        </menu>
    </item>
</menu>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

</LinearLayout>
```

MainActivity.java

```
package com.example.submenuapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    // Create Menu
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu_main, menu);
        return true;
    }

    // Handle Sub Menu Click
    @Override
    public boolean onOptionsItemSelected(MenuItem item) {

        int id = item.getItemId();

        if(id == R.id.profile){
            Toast.makeText(this, "Profile Selected", Toast.LENGTH_SHORT).show();
        }
        else if(id == R.id.settings){
            Toast.makeText(this, "Settings Selected", Toast.LENGTH_SHORT).show();
        }
        else if(id == R.id.logout){
            Toast.makeText(this, "Logout Selected", Toast.LENGTH_SHORT).show();
```

```
        }

        return super.onOptionsItemSelected(item);
    }

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Design a screen with two buttons: Start and Stop. When the Start button is clicked, start a Timer (display the time in seconds). When the Stop button is clicked, stop the timer.

MainActivity.java

```
package com.example.timerapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.os.Handler;
import android.widget.Button;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    TextView textTimer;
    Button startBtn, stopBtn;

    Handler handler = new Handler();
    int seconds = 0;
    boolean running = false;

    Runnable runnable = new Runnable() {
        @Override
        public void run() {

            if(running){
                seconds++;
                textTimer.setText(String.valueOf(seconds));
                handler.postDelayed(this, 1000); // repeat every 1 sec
            }
        }
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```
        textTimer = findViewById(R.id.textTimer);
        startBtn = findViewById(R.id.buttonStart);
        stopBtn = findViewById(R.id.buttonStop);

        startBtn.setOnClickListener(v -> {

            if(!running){
                running = true;
                handler.post(runnable);
            }
        });

        stopBtn.setOnClickListener(v -> running = false);
    }
}
```

activity_main

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:gravity="center"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <!-- Timer Display -->
    <TextView
        android:id="@+id/textTimer"
        android:text="0"
        android:textSize="40sp"
        android:layout_marginBottom="30dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <!-- Start Button -->
    <Button
        android:id="@+id/buttonStart"
        android:text="Start"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <!-- Stop Button -->
    <Button
        android:id="@+id/buttonStop"
        android:text="Stop"
        android:layout_marginTop="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

Create a simple Registration Form with the following fields:

Name (Text Input)
Email (Text Input)
Phone Number (Text Input)
Password (Password Input)
Confirm Password (Password Input)
Submit Button

activity_main

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent">
   <LinearLayout
       android:orientation="vertical"
       android:padding="24dp"
       android:layout_width="match_parent"
       android:layout_height="wrap_content">
       <EditText
           android:hint="@string/name"
           android:layout_width="match_parent"
           android:layout_height="wrap_content"/>
       <EditText
           android:hint="@string/email"
           android:inputType="textEmailAddress"
           android:layout_marginTop="12dp"
           android:layout_width="match_parent"
           android:layout_height="wrap_content"/>
       <EditText
           android:hint="@string/phone"
           android:inputType="phone"
           android:layout_marginTop="12dp"
           android:layout_width="match_parent"
           android:layout_height="wrap_content"/>
       <EditText
           android:hint="@string/password"
           android:inputType="textPassword"
           android:layout_marginTop="12dp"
           android:layout_width="match_parent"
           android:layout_height="wrap_content"/>
 <EditText
           android:hint="@string/confirm_password"
           android:inputType="textPassword"
           android:layout_marginTop="12dp"
           android:layout_width="match_parent"
           android:layout_height="wrap_content"/>

       <!-- Submit Button -->
       <Button
           android:text="@string/submit"
```

```
            android:layout_marginTop="20dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

    </LinearLayout>

</ScrollView>
```

strings.xml

```
<resources>
    <string name="app_name">RegistrationFormApp</string>
    <string name="name">Name</string>
    <string name="email">Email</string>
    <string name="phone">Phone Number</string>
    <string name="password">Password</string>
    <string name="confirm_password">Confirm Password</string>
    <string name="submit">Submit</string>
</resources>
```

MainActivity.java

```
package com.example.registrationformapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Create a Registration Form with the following fields:

- **First Name (Text Input)**
- **Last Name (Text Input)**
- **Email Address (Text Input)**
- **Country (Spinner Dropdown)**
- **Password (Password Input)**
- **Confirm Password (Password Input)**
- **Submit Button**

MainActivity.java

```
package com.example.registrationformspinner;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
```

```java
import android.widget.ArrayAdapter;
import android.widget.Spinner;

public class MainActivity extends AppCompatActivity {

   Spinner spinnerCountry;

   String[] countries = {
           "Select Country",
           "India",
           "USA",
           "UK",
           "Canada",
           "Australia"
   };

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       spinnerCountry = findViewById(R.id.spinnerCountry);

       ArrayAdapter<String> adapter = new ArrayAdapter<>(
               this,
               android.R.layout.simple_spinner_item,
               countries
       );

       adapter.setDropDownViewResource(
               android.R.layout.simple_spinner_dropdown_item
       );

       spinnerCountry.setAdapter(adapter);
   }
}
```

Activity_main.xml

```java
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <LinearLayout
       android:orientation="vertical"
       android:padding="24dp"
       android:layout_width="match_parent"
       android:layout_height="wrap_content">

       <!-- First Name -->
```

```
<EditText
    android:id="@+id/editFirstName"
    android:hint="@string/first_name"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<!-- Last Name -->
<EditText
    android:id="@+id/editLastName"
    android:hint="@string/last_name"
    android:layout_marginTop="12dp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<!-- Email -->
<EditText
    android:id="@+id/editEmail"
    android:hint="@string/email"
    android:inputType="textEmailAddress"
    android:layout_marginTop="12dp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<!-- Country Spinner -->
<Spinner
    android:id="@+id/spinnerCountry"
    android:layout_marginTop="12dp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<!-- Password -->
<EditText
    android:id="@+id/editPassword"
    android:hint="@string/password"
    android:inputType="textPassword"
    android:layout_marginTop="12dp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<!-- Confirm Password -->
<EditText
    android:id="@+id/editConfirmPassword"
    android:hint="@string/confirm_password"
    android:inputType="textPassword"
    android:layout_marginTop="12dp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<!-- Submit Button -->
```

```
        <Button
            android:id="@+id/buttonSubmit"
            android:text="@string/submit"
            android:layout_marginTop="20dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

    </LinearLayout>

</ScrollView>
```

Strings.xml

```
<resources>
    <string name="app_name">RegistrationFormSpinner</string>
    <string name="first_name">First Name</string>
    <string name="last_name">Last Name</string>
    <string name="email">Email Address</string>
    <string name="password">Password</string>
    <string name="confirm_password">Confirm Password</string>
    <string name="submit">Submit</string>
</resources>
```

Create a Registration Form with the following fields:

Username (Text Input)
Email Address (Text Input)
Gender (Radio Buttons with options ;Male and Female
Password (Password Input)
Submit Button

Project Name: RegistrationRadioApp

Mainactivity.java

```
package com.example.registrationradioapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Button;
import android.widget.RadioGroup;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    RadioGroup radioGroupGender;
    Button buttonSubmit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);

        radioGroupGender = findViewById(R.id.radioGroupGender);
        buttonSubmit = findViewById(R.id.buttonSubmit);
        buttonSubmit.setOnClickListener(v -> {
            int selectedId = radioGroupGender.getCheckedRadioButtonId();

            if (selectedId == R.id.radioMale) {
                Toast.makeText(this, "Gender: Male", Toast.LENGTH_SHORT).show();
            } else if (selectedId == R.id.radioFemale) {
                Toast.makeText(this, "Gender: Female", Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "Please select Gender",
Toast.LENGTH_SHORT).show();
            }

        });
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:orientation="vertical"
        android:padding="24dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <!-- Username -->
        <EditText
            android:id="@+id/editUsername"
            android:hint="@string/username"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <!-- Email -->
        <EditText
            android:id="@+id/editEmail"
            android:hint="@string/email"
            android:inputType="textEmailAddress"
            android:layout_marginTop="12dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <TextView
            android:text="@string/gender"
```

```
            android:layout_marginTop="16dp"
            android:textSize="16sp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>
        <RadioGroup
            android:id="@+id/radioGroupGender"
            android:orientation="horizontal"
            android:layout_marginTop="8dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content">

            <RadioButton
                android:id="@+id/radioMale"
                android:text="@string/male"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"/>

            <RadioButton
                android:id="@+id/radioFemale"
                android:text="@string/female"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"/>

        </RadioGroup>

        <!-- Password -->
        <EditText
            android:id="@+id/editPassword"
            android:hint="@string/password"
            android:inputType="textPassword"
            android:layout_marginTop="12dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <!-- Submit Button -->
        <Button
            android:id="@+id/buttonSubmit"
            android:text="@string/submit"
            android:layout_marginTop="20dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

    </LinearLayout>

</ScrollView>
```

Strings.xml

```
<resources>
    <string name="app_name">RegistrationRadioApp</string>
```

```
    <string name="username">Username</string>
    <string name="email">Email Address</string>
    <string name="gender">Gender</string>
    <string name="male">Male</string>
    <string name="female">Female</string>
    <string name="password">Password</string>
    <string name="submit">Submit</string>
</resources>
```

Create a Registration Form with the following fields:

- Username (Text Input)
- Email Address (Text Input)
- Phone Number (Text Input)
- Password (Password Input)
- Submit Button (Floating Action Button)

Project Name: RegistrationFABApp

MainActivity.java

```
package com.example.registrationfabapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Toast;
import com.google.android.material.floatingactionbutton.FloatingActionButton;

public class MainActivity extends AppCompatActivity {

    FloatingActionButton fabSubmit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        fabSubmit = findViewById(R.id.fabSubmit);

        fabSubmit.setOnClickListener(v ->
                Toast.makeText(this, "Form Submitted", Toast.LENGTH_SHORT).show()
        );
    }
}
```

Activity_Main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
android:padding="24dp">

<!-- Username -->
<EditText
    android:id="@+id/editUsername"
    android:hint="@string/username"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"/>

<!-- Email -->
<EditText
    android:id="@+id/editEmail"
    android:hint="@string/email"
    android:inputType="textEmailAddress"
    android:layout_marginTop="12dp"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    app:layout_constraintTop_toBottomOf="@id/editUsername"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"/>

<!-- Phone -->
<EditText
    android:id="@+id/editPhone"
    android:hint="@string/phone"
    android:inputType="phone"
    android:layout_marginTop="12dp"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    app:layout_constraintTop_toBottomOf="@id/editEmail"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"/>

<!-- Password -->
<EditText
    android:id="@+id/editPassword"
    android:hint="@string/password"
    android:inputType="textPassword"
    android:layout_marginTop="12dp"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    app:layout_constraintTop_toBottomOf="@id/editPhone"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"/>

<com.google.android.material.floatingactionbutton.FloatingActionButton
```

```
        android:id="@+id/fabSubmit"
        android:contentDescription="@string/submit"
        app:srcCompat="@android:drawable/ic_menu_send"
        android:layout_margin="16dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

Strings.xml

```
<resources>
   <string name="app_name">RegistrationFABApp</string>
   <string name="username">Username</string>
   <string name="email">Email Address</string>
   <string name="phone">Phone Number</string>
   <string name="password">Password</string>
   <string name="submit">Submit</string>
</resources>
```

Design a Registration Form that includes:

- Name (Text Input)
- Date of Birth (Date Picker)
- Email Address (Text Input)
- Phone Number (Text Input)
- Password (Password Input)
- Submit Button

Name: RegistrationDatePickerApp
Language: Java
Minimum SDK: API 21+

Mainactivity.java

```
package com.example.registrationdatepickerapp;

import androidx.appcompat.app.AppCompatActivity;
import android.app.DatePickerDialog;
import android.os.Bundle;
import android.widget.EditText;

import java.util.Calendar;

public class MainActivity extends AppCompatActivity {
```

```
    EditText editDOB;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editDOB = findViewById(R.id.editDOB);

        editDOB.setOnClickListener(v -> {

            Calendar calendar = Calendar.getInstance();
            int year = calendar.get(Calendar.YEAR);
            int month = calendar.get(Calendar.MONTH);
            int day = calendar.get(Calendar.DAY_OF_MONTH);

            DatePickerDialog datePickerDialog =
                    new DatePickerDialog(MainActivity.this,
                            (view, selectedYear, selectedMonth, selectedDay) -> {

                                String date = selectedDay + "/" +
                                        (selectedMonth + 1) + "/" +
                                        selectedYear;

                                editDOB.setText(date);
                            },
                            year, month, day);

            datePickerDialog.show();
        });
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:orientation="vertical"
        android:padding="24dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <EditText
            android:id="@+id/editName"
            android:hint="@string/name"
```

```
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <EditText
            android:id="@+id/editDOB"
            android:hint="@string/dob"
            android:focusable="false"
            android:clickable="true"
            android:layout_marginTop="12dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <EditText
            android:id="@+id/editEmail"
            android:hint="@string/email"
            android:inputType="textEmailAddress"
            android:layout_marginTop="12dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <EditText
            android:id="@+id/editPhone"
            android:hint="@string/phone"
            android:inputType="phone"
            android:layout_marginTop="12dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <EditText
            android:id="@+id/editPassword"
            android:hint="@string/password"
            android:inputType="textPassword"
            android:layout_marginTop="12dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <Button
            android:id="@+id/buttonSubmit"
            android:text="@string/submit"
            android:layout_marginTop="20dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

    </LinearLayout>

</ScrollView>
```

Strings.xml

```
<resources>
```

```
    <string name="app_name">RegistrationDatePickerApp</string>
    <string name="name">Name</string>
    <string name="dob">Date of Birth</string>
    <string name="email">Email Address</string>
    <string name="phone">Phone Number</string>
    <string name="password">Password</string>
    <string name="submit">Submit</string>
</resources>
```

Create a Registration Form with the following fields:

- Username (Text Input)
- Email Address (Text Input)
- Agree to Terms and Conditions (Checkbox)
- Password (Password Input)
- Confirm Password (Password Input)
- Submit Button

Name: RegistrationCheckboxApp
Language: Java
Minimum SDK: API 21+

MainActivity.java

```
package com.example.registrationcheckboxapp;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    CheckBox checkTerms;
    Button buttonSubmit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        checkTerms = findViewById(R.id.checkTerms);
        buttonSubmit = findViewById(R.id.buttonSubmit);

        buttonSubmit.setOnClickListener(v -> {

            if (checkTerms.isChecked()) {
```

```
                Toast.makeText(this,
                        "Registration Successful",
                        Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this,
                        "Please accept Terms and Conditions",
                        Toast.LENGTH_SHORT).show();
            }

        });
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <LinearLayout
       android:orientation="vertical"
       android:padding="24dp"
       android:layout_width="match_parent"
       android:layout_height="wrap_content">

       <!-- Username -->
       <EditText
           android:id="@+id/editUsername"
           android:hint="@string/username"
           android:layout_width="match_parent"
           android:layout_height="wrap_content"/>

       <EditText
           android:id="@+id/editEmail"
           android:hint="@string/email"
           android:inputType="textEmailAddress"
           android:layout_marginTop="12dp"
           android:layout_width="match_parent"
           android:layout_height="wrap_content"/>
       <CheckBox
           android:id="@+id/checkTerms"
           android:text="@string/terms"
           android:layout_marginTop="12dp"
           android:layout_width="wrap_content"
           android:layout_height="wrap_content"/
       <EditText
           android:id="@+id/editPassword"
           android:hint="@string/password"
           android:inputType="textPassword"
```

```
            android:layout_marginTop="12dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>
        <EditText
            android:id="@+id/editConfirmPassword"
            android:hint="@string/confirm_password"
            android:inputType="textPassword"
            android:layout_marginTop="12dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>
        <Button
            android:id="@+id/buttonSubmit"
            android:text="@string/submit"
            android:layout_marginTop="20dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>
    </LinearLayout>
</ScrollView>
```

Strings.xml

```
<resources>
    <string name="app_name">RegistrationCheckboxApp</string>
    <string name="username">Username</string>
    <string name="email">Email Address</string>
    <string name="terms">Agree to Terms and Conditions</string>
    <string name="password">Password</string>
    <string name="confirm_password">Confirm Password</string>
    <string name="submit">Submit</string>
</resources>
```

Create a Registration Form with the following fields:

- Username (Text Input)
- Email Address (Text Input)
- Password (Password Input)
- Confirm Password (Password Input)
- Profile Picture (Image Upload Button)
- Submit Button

Name: RegistrationImageUploadApp
Language: Java
Minimum SDK: API 21+

MainActivity.java

```
package com.example.registrationimageuploadapp;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.net.Uri;
```

```java
import android.os.Bundle;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

   ImageView imageProfile;
   Button buttonUpload, buttonSubmit;

   private static final int PICK_IMAGE = 1;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       imageProfile = findViewById(R.id.imageProfile);
       buttonUpload = findViewById(R.id.buttonUpload);
       buttonSubmit = findViewById(R.id.buttonSubmit);

       buttonUpload.setOnClickListener(v -> {

           Intent intent = new Intent();
           intent.setType("image/*");
           intent.setAction(Intent.ACTION_GET_CONTENT);

           startActivityForResult(
                   Intent.createChooser(intent, "Select Picture"),
                   PICK_IMAGE
           );
       });

       buttonSubmit.setOnClickListener(v ->
               Toast.makeText(this,
                       "Registration Submitted",
                       Toast.LENGTH_SHORT).show());
   }

   @Override
   protected void onActivityResult(int requestCode, int resultCode, Intent data) {

       super.onActivityResult(requestCode, resultCode, data);

       if (requestCode == PICK_IMAGE &&
               resultCode == RESULT_OK &&
               data != null) {

           Uri imageUri = data.getData();
```

```
            imageProfile.setImageURI(imageUri);
        }
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <LinearLayout
       android:orientation="vertical"
       android:padding="24dp"
       android:layout_width="match_parent"
       android:layout_height="wrap_content">

       <!-- Username -->
       <EditText
           android:id="@+id/editUsername"
           android:hint="@string/username"
           android:layout_width="match_parent"
           android:layout_height="wrap_content"/>

       <!-- Email -->
       <EditText
           android:id="@+id/editEmail"
           android:hint="@string/email"
           android:inputType="textEmailAddress"
           android:layout_marginTop="12dp"
           android:layout_width="match_parent"
           android:layout_height="wrap_content"/>

       <!-- Password -->
       <EditText
           android:id="@+id/editPassword"
           android:hint="@string/password"
           android:inputType="textPassword"
           android:layout_marginTop="12dp"
           android:layout_width="match_parent"
           android:layout_height="wrap_content"/>

       <!-- Confirm Password -->
       <EditText
           android:id="@+id/editConfirmPassword"
           android:hint="@string/confirm_password"
           android:inputType="textPassword"
           android:layout_marginTop="12dp"
           android:layout_width="match_parent"
```

```
            android:layout_height="wrap_content"/>

        <!-- Profile Image -->
        <ImageView
            android:id="@+id/imageProfile"
            android:layout_width="120dp"
            android:layout_height="120dp"
            android:src="@android:drawable/ic_menu_gallery"
            android:layout_gravity="center"
            android:layout_marginTop="16dp"/>

        <!-- Upload Button -->
        <Button
            android:id="@+id/buttonUpload"
            android:text="@string/upload"
            android:layout_marginTop="12dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <!-- Submit Button -->
        <Button
            android:id="@+id/buttonSubmit"
            android:text="@string/submit"
            android:layout_marginTop="20dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

    </LinearLayout>

</ScrollView>
```

Strings.xml

```
<resources>
    <string name="app_name">RegistrationImageUploadApp</string>
    <string name="username">Username</string>
    <string name="email">Email Address</string>
    <string name="password">Password</string>
    <string name="confirm_password">Confirm Password</string>
    <string name="upload">Upload Profile Picture</string>
    <string name="submit">Submit</string>
</resources>
```

Create a BroadcastReceiver that listens for the Android Intent ACTION_BATTERY_CHANGED broadcast and shows a Toast displaying the current battery percentage.

Name : batteryreceiverapp

Mainactivity.java

```
package com.example.batteryreceiverapp;

import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;
import android.content.IntentFilter;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

   BatteryReceiver batteryReceiver;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       batteryReceiver = new BatteryReceiver();
   }

   @Override
   protected void onResume() {
       super.onResume();

       IntentFilter filter =
               new IntentFilter(Intent.ACTION_BATTERY_CHANGED);

       registerReceiver(batteryReceiver, filter);
   }

   @Override
   protected void onPause() {
       super.onPause();

       unregisterReceiver(batteryReceiver);
   }
}
```

Batteryreceiver.java

```
package com.example.batteryreceiverapp;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.os.BatteryManager;
import android.widget.Toast;

public class BatteryReceiver extends BroadcastReceiver {
```

```
    @Override
    public void onReceive(Context context, Intent intent) {

        int level = intent.getIntExtra(BatteryManager.EXTRA_LEVEL, -1);
        int scale = intent.getIntExtra(BatteryManager.EXTRA_SCALE, -1);

        int batteryPercentage = (int) ((level / (float) scale) * 100);

        Toast.makeText(context,
                "Battery Level: " + batteryPercentage + "%",
                Toast.LENGTH_SHORT).show();
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:text="Battery Receiver Running..."
        android:textSize="18sp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

Create an app that listens for ACTION_AIRPLANE_MODE_CHANGED using a BroadcastReceiver. Display a Toast message whenever the airplane mode is turned on or off.